Venkata Lakshmi Parimala Pasupuleti

Prompt Engineer · RAG Pipelines · LLM Evaluation · Guardrails & PII Compliance · Document Intelligence · Python

venkatalakshmiparimala@gmail.com | linkedin.com/in/venkata-lakshmi-parimala-pasupuleti | LeaseI (live) | AgentProbe (live) | Virginia, USA

PROFESSIONAL SUMMARY

Prompt engineer and AI systems builder with 8+ years of production engineering in regulated environments (federal housing compliance: LIHTC, HUD, HAP, AMI — zero violations). Built **LeaseI**: a RAG platform over lease and compliance documents with OCR-fallback ingestion, section-aware chunking, 100% citation correctness, hallucination mitigation enforced in architecture, and a hard safety boundary against ungrounded AI responses. Built **AgentProbe**: a CI-integrated evaluation harness measuring factual accuracy, hallucination, retrieval quality, completeness, and compliance on every commit — regression-gated before merge. Expert at translating business problems across Operations, Legal, and cross-functional teams into production AI solutions. M.S. Data Analytics Engineering, GMU (GPA 3.87, May 2026).

CORE COMPETENCIES

Prompt Engineering (Chain-of-Thought, Few-Shot, Zero-Shot, Role Spec, XML) | RAG Pipeline Design | LLM Evaluation Harnesses | Hallucination Mitigation | PII Handling & PHI Compliance | Guardrails & Refusal Behavior | OCR & Document Intelligence | Regulated Domain AI (Federal Compliance) | Cross-Functional AI Enablement | Claude & OpenAI APIs | Production Python | Vector Databases (ChromaDB)

EXPERIENCE

Magizhs Tech Solution

May 2025 – Aug 2025

AI/ML Engineer — Prompt Engineering · RAG · Guardrails · Regulated Compliance

- Designed and shipped **production prompt workflows** for marketing, real estate, and commercial survey clients — system prompts with chain-of-thought reasoning, structured output schemas, and deterministic guardrails enforced before any action fires; compliant outputs route to export, non-compliant ones route to rejection with grounded regulatory citations.
- Built a **RAG pipeline over FAA Part 107 corpus** with cited-source compliance checks — every answer grounded in source regulation; zero unsupported claims in production; confidence-aware routing determines when to respond vs. escalate to human review.

George Mason University — Research Associate / Teaching Assistant

Aug 2024 – May 2026

Research · Data Analysis · Python Instruction

- Built statistical prediction models (Python, R) achieving $R^2 = 0.977$ across 190+ countries; applied Isolation Forest anomaly detection surfacing data integrity issues manual review missed; mentored 600+ students in Python programming and database design.

Entrata (PropTech / Real Estate SaaS)

Sep 2018 – Aug 2024

Senior Software Engineer — Federal Compliance · PHI & PII Security · AI Pipeline · Cross-Functional Delivery

- **Shipped Entrata's first production AI pipeline** across 1.6M real financial records (Python, PyTorch, OpenAI API) — automated data processing workflows with output validation, quality gates, and compliance controls enforced from day one; zero violations at launch.
- **Achieved 100% PHI compliance** on AWS S3 migration — encrypted 100% of resident data (phone numbers, SSNs) using Libsodium; updated 50%+ of product codebase for encryption compliance; zero security incidents; direct experience embedding PII handling rules into production systems.
- Built and shipped **20+ production modules** (Python, React, PHP, PostgreSQL, RabbitMQ) for 100K+ monthly active users across federal housing compliance programs (LIHTC, HUD, HAP, AMI, Section 8/HCV) — zero compliance violations; 99.5% uptime across 6+ months.
- **Cross-functional delivery across 120+ Agile sprints**: translated federal housing requirements into production architecture across Compliance, Operations, Product, and Legal stakeholders — the same cross-team translation this role requires (Operations, Control Tower, General Counsel).
- Eliminated 80% of manual operational workload via Python/Bash automation pipelines; mentored 5+ engineers; conducted 300+ peer code reviews.

Entrata India (Db Xento Systems)

Jan 2017 – Sep 2018

Associate / Trainee Software Engineer — Backend · Data Pipelines · Compliance Modules

- Built backend modules for affordable housing compliance workflows; processed 1.6M HUD/Tax Credit records; created 56 state-specific certification documents with merge fields; migrated client documents to AWS S3.

PORTFOLIO PROJECTS

Leasel — RAG Platform with OCR, Hallucination Mitigation & Eval Harness

Prompt Engineering | RAG | Guardrails

2026 · Python, LangGraph, FastAPI, TF-IDF, OCR, SQLite · github.com/venkatalakshmiparimala/Lease_complaince_rag

- Built a RAG platform over regulatory documents with **OCR-fallback ingestion** (Tesseract for scanned PDFs), section-aware chunking, TF-IDF retrieval, and citation-grounded output — 100% citation correctness, 0% unsupported-claim rate; core rule enforced in code: every answer cited to an exact source clause or flagged as unsupported and downgraded.
- **Hallucination mitigation enforced in architecture**: LLM extracts and reasons; deterministic logic or human review makes every regulated eligibility call — never the reverse. 0 safety boundary violations on full validation set; tested for prompt injection and adversarial inputs in tenant-facing surface.
- Achieved **93.6% precision and 87.9% F1** across 10 compliance rules on 1,240 lease PDFs (5 US states + federal); LLM-as-judge evaluation with multi-trial sampling and regression gate — same evaluation harness structure the JD requires.

AgentProbe — CI-Integrated LLM Evaluation & Prompt Benchmarking Platform

LLM Eval | Prompt Benchmarking | Regression Gate

2024–Present · Python, LangGraph, FastAPI, ChromaDB, Anthropic, OpenAI, Groq, Ollama, React, Docker, GitHub Actions · github.com/venkatalakshmiparimala/AgentProbe

- Built a **CI-integrated evaluation harness** measuring 5 metrics on every commit: factual accuracy, hallucination, retrieval quality, completeness, and compliance — regressions blocked before merge; per-run benchmarks tracked over time; directly analogous to the evaluation harness this role requires for prompt quality, consistency, and regression over model updates.
- Multi-provider LLM routing (Anthropic, OpenAI, Groq, Ollama) enables **prompt variant testing across providers** — compare prompt performance across models, track A/B benchmark results per run, detect drift in retrieval quality and response consistency over model updates.

EDUCATION

M.S. Data Analytics Engineering — George Mason University, Virginia — May 2026 · GPA 3.87

B.Tech Computer Science & Engineering — JNTUA, India — 2013–2017

TECHNICAL SKILLS

Prompt Engineering: Prompt Engineering, Chain-of-Thought Prompting, Few-Shot Prompting, Zero-Shot Prompting, Role Specification, System Prompt Design, XML-Structured Inputs, Output Validation, Hallucination Mitigation, Hallucination Detection, Guardrails, Refusal Behavior, Prompt Injection Testing, Prompt Variant Testing, Context Engineering, Spec Engineering

RAG & LLM Evaluation: RAG Pipelines, Semantic Search, Embeddings, Vector Database, ChromaDB, TF-IDF Retrieval, LLM Evaluation, LLM-as-Judge, Eval Frameworks, CI-Integrated Evals, Regression Gating, Anthropic API, OpenAI API, Groq, Ollama, Multi-Provider LLM Routing, LangChain, LangGraph, FastAPI

Document Intelligence & Compliance: OCR (Tesseract), Document Intelligence, Section-Aware Chunking, Citation Grounding, PII Handling, PHI Compliance (Libsodium), Regulated Environment AI, Federal Compliance (LIHTC, HUD, HAP, AMI)

Backend & Infrastructure: Python (primary), React, JavaScript, TypeScript, SQL, AWS (S3, EC2, RDS, Glue), Docker, CI/CD (GitHub Actions), PostgreSQL, RabbitMQ, Microservices, ETL, Git, Linux, Agile/Scrum